Evaluation instrument choice can flip the apparent winner: a secondary analysis reconciling two contradictory 2026 head-to-head evaluations of clinical AI

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Abstract

Background. Two 2026 head-to-head studies reached opposite conclusions about specialized clinical AI. Real-POCQi (arXiv:2606.28960) found OpenEvidence (OE) beat frontier general-purpose LLMs on blinded physician **pairwise preference** across 620 real point-of-care queries; a Nature Medicine study (s41591-026-04431-5) found frontier LLMs beat OE and UpToDate Expert AI under **absolute 1–4 rubric** scoring. Each study also sourced its "real-world" queries from the platform that won (home-field provenance), and the two used different model versions and evaluation instruments, confounding any direct comparison.

Objective. Test whether the **evaluation instrument alone** — with queries and answers held fixed — can flip the winner, isolating it from query provenance, model version, and answer content.

Methods. We reused Real-POCQi's public data (CC BY 4.0): the same 620 queries, the same four systems' verbatim answers (Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence), and the same blinded human pairwise ratings. On a seed-fixed 150-query sample we re-scored every answer with a **Nature-style absolute 1–4 rubric** administered by a blinded four-model LLM-judge panel (GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash), all at **high reasoning effort with token consumption verified**. We re-expressed both instruments through the same OE-vs-rest win-difference summary (a construct re-expression, not an identity — the rubric win-difference is a thresholded score gap and is hypersensitive at the decision boundary) and compared them, with a **crossed question × judge bootstrap that treats the four judges as a random factor** (so CIs reflect panel composition, not just question sampling), per-judge self-preference, and leave-one-judge-out robustness. To separate the *instrument* effect from a *rater-population* effect (physicians vs LLMs), we additionally ran the same blinded panel under the **pairwise** instrument, filling three of the four cells of a {pairwise, rubric} × {human, LLM} design and decomposing the swing into rater-modality (B–A) and instrument-format (C–B) components with propagated CIs. A restricted version of this pairwise-LLM cell already exists inside Real-POCQi, so we frame it as replication-and-extension.

Results. Our **primary estimand holds the rater fixed and varies only the format**: for each of the four judges we threshold that judge's own rubric scores into win/tie/loss and pool them exactly as its pairwise votes, then average the per-judge pairwise→rubric change (an aggregation-matched, same-judge format component). This component is **negative on all five axes** — accuracy −16.9, clinical utility −21.3, source quality −40.6, completeness −37.4, verifiability −45.8 pp — **sign-consistent across all four judges**, Holm-significant, with simultaneous max-|T| confidence intervals excluding zero, and robust both to resampling the judge panel and to restricting the analysis to OE answers with no detected citation markers. Changing only the evaluation format — same answers, same rater — thus moves and often reverses the OE-vs-frontier ranking. The effect is **largest and most robust on the evidence-presentation axes** (source quality, verifiability) and **smallest, and aggregation-sensitive, on accuracy**: the earlier −29.1 pp accuracy figure came from averaging the four judges and *then* thresholding, which more than doubles a small

native gap (individual-matched −13.0 pp; a quarter-point tie-band on the panel measure gives −13.8 pp; the native OE–frontier accuracy gap is only −0.125 on the 1–4 scale). We therefore demote accuracy from marquee axis to the **weakest** instance of the effect. On the **exact common support** (matching every question×opponent×axis with a human vote *and* complete four-judge pairwise and rubric observations; 108 question×opponent units/axis) the format component (C–B) is negative on all five axes (−17.8 to −43.1, CIs exclude zero), while the **human→LLM protocol contrast (B–A)** — which changes rater population **and** answer rendering (physicians saw text-only; LLM judges saw citation-bearing Markdown) — is **null on accuracy (−7.9 pp, CI −23.5 to +7.1, bootstrap tail p=0.30)** and clinical utility but **significantly positive** on source quality (+25.1), completeness (+17.6) and verifiability (+30.0). We do not attribute this to rater population alone: the positive contrasts fall precisely on the evidence-presentation axes most sensitive to seeing citations, so B–A confounds rater with rendering. (This also corrects a prior-revision error that reported an accuracy B–A of −24.0 pp by inadvertently comparing full-bank human ratings against matched LLM ratings; on matched support it is small and non-significant.) The decomposition is robust to **equal-key weighting** (each question×opponent counted once): accuracy B–A −5.8 (still NS), evidence axes +18.8 to +29.5 (still significant), format component unchanged. GPT-5.5 self-preference survives as a difference-in-differences (+0.42 beyond panel consensus), and length does not drive the gap (OE is the *longest* provider yet loses accuracy).

Conclusions. Within Real-POCQi, **both** the evaluation format (pairwise vs absolute rubric) **and** the human→LLM protocol contrast (rater population *and* answer rendering, plus specialty-matching) can materially alter the apparent ranking — but they act differently: the format component is negative on all five axes, whereas the protocol contrast is null on accuracy and significantly *positive* on the evidence axes. These are two plausible contributors to the disagreement between the two published studies, which also differ in datasets, provenance, model versions, access paths, exact rubrics, and human populations; we do **not** claim a complete causal decomposition of that disagreement, and we no longer claim the format acts "rather than" the rater. The robust, assumption-light result holds the rater fixed: **with the same LLM judge, changing only the format significantly shifts and often reverses the OE-vs-frontier ranking on all five axes.** This is a **three-cell path decomposition** ({pairwise, rubric} × {human, LLM} with the human-rubric cell missing), not a completed factorial: we identify the human→LLM change under pairwise and the pairwise→rubric change within LLMs, but neither the format effect among humans nor the rater×format interaction — and the missing cell is exactly where an interaction is most plausible. Bridging the within-LLM format effect to the human-rated Nature rubric therefore assumes that interaction is small, which we flag as the central untested step. A mechanism is visible in the data: the absolute rubric is ceiling-bound on content axes (72–82% top score) but discriminating on evidence axes, where a forced pairwise choice rewards the citations and clinician-tuned framing that rubric scoring compresses. Subject to these limits, "which clinical AI is best" is not format-invariant; leaderboards should report the evaluation format as a first-class experimental factor. We do **not** reproduce Nature's exact RCQ rubric (we score the five Real-POCQi axes with an LLM panel), so this is a general absolute-vs-comparative contrast, not a transport of the Nature instrument. We additionally document two under-reported flaws common to both source studies — omission of the actual ChatGPT product clinicians use, and unreported/uncontrolled reasoning effort — and pre-register a full provenance × instrument × citation factorial (including the missing human-rubric cell).

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1. Introduction

Point-of-care clinical AI is now evaluated by head-to-head studies whose results directly influence purchasing and deployment. Medical-AI evaluation has grown rapidly, spanning licensing-exam question banks,[1] physician-aligned rubric benchmarks,[2] large language models that encode clinical knowledge,[3] specialized and mobile-sized clinical models,[4,5,6] health-system-scale prediction and operations models,[7,8] sequential diagnostic reasoning,[9] retrieval-augmented generation for healthcare,[10] and a parallel literature documenting these systems' failure modes — susceptibility to distraction,[11] conflict between an LLM's internal prior and retrieved evidence,[12] data-poisoning attacks,[13] and the need for clinically safe generation.[14] These benchmarks increasingly characterize their query distributions with topic models.[15] Deployment is already at health-system scale.[16] In 2026 two such studies reached **opposite** conclusions using overlapping systems,[17,18] which strongly suggests that at least one conclusion reflects the evaluation method rather than the systems compared.

Both studies contain a symmetric structural feature — each sourced its "real-world" query set from the platform that ultimately won (Real-POCQi from OpenEvidence traffic; the Nature study's real-clinical- query benchmark from an NYU Langone GPT deployment) — and each used a **different evaluation instrument** (blinded human pairwise preference vs absolute rubric scoring). Provenance and instrument are therefore fully confounded in the published record. This paper isolates the instrument by holding everything else constant, and then situates that result within a fuller critique and a pre-registered factorial design.

Our contributions: 1. **A format effect with the rater held fixed, plus a three-cell path decomposition** (executed, on public data): holding queries, answers, **and the rater** fixed, changing only the evaluation format shifts or reverses OE's advantage on all five axes — the aggregation-matched, same-judge format component is negative and sign-consistent across all four judges (§4.1, §4.5a). Filling three cells of a {pairwise, rubric} × {human, LLM} design, on **exact common support**, shows the format component (C−B) is negative on all five axes while the human→LLM rater component (B−A) is null on accuracy and significantly *positive* on the evidence axes (§4.5b). So we do **not** claim the format acts *instead of* the rater — both contribute — but the reversal is carried by the format; the rater does not explain it. 2. **Two methodological critiques** of both source studies: the comparator set omits the ChatGPT product clinicians actually use, and reasoning effort is unreported or uncontrolled (§5). 3. **A pre-registered 2×2×2** provenance × instrument × citation design (§7) that would estimate each factor's causal contribution, with power grounded in measured variance components. This is a *proposed design, not an executed result* — we flag it as such to keep the executed contribution (#1) cleanly separated from future work.

2. The two source studies (verified)

Numbers below were verified for this work from primary sources (Nature PDF Methods; Real-POCQi abstract + HTML full text).

Real-POCQi[17] (arXiv:2606.28960; dataset `jjfenglab/Real-POCQi`, CC BY 4.0). 620 real point-of-care queries sourced from OpenEvidence[19] plus 187 HealthBench[2] items; 149 physicians across 36 states; blinded **pairwise** comparisons — the arena-style preference paradigm[20] — on five axes (accuracy, clinical utility, source quality, completeness, verifiability). Systems: Claude Opus 4.8, Gemini 3.1 Pro, GPT-5.5, OpenEvidence, all queried via API; temperature 0.0, seed 42, web search enabled; **"Thinking was automatically determined by the LLM."** OE won by roughly +25 to +39 pp win-difference. *The data collection was run by the platform under study (conflict of interest).*

Nature Medicine[18] (s41591-026-04431-5, NYU Langone + UT Austin). MedQA[1] (500) + HealthBench[2] (500) + a 100-item real-clinical-question (RCQ) benchmark sampled from NYU's HIPAA-compliant GPT instance;[16] 12 clinician raters; **absolute 1–4 rubric**. Systems: GPT-5.2 (2025-12-11), Gemini 3.1 Pro Preview, Claude Opus 4.6 (all API), OpenEvidence and UpToDate Expert AI[21] (browser), plus Google Search AI Overview (RCQ). Temperature 0.0, seed 62, search enabled; **reasoning effort not reported.** Length not normalized (by explicit choice). Frontier LLMs won: e.g. HealthBench GPT 88.0 / Gemini 79.3 / Claude 77.0 / OE 62.6 / UpToDate 61.3; RCQ Gemini 3.62 / GPT 3.54 / Claude 3.52 / OE 3.24. HealthBench was graded by an LLM-judge panel (self-preference risk); RCQ item-level agreement was low (Krippendorff $\alpha \approx$ 0.10–0.20). License CC BY-NC-ND 4.0.

The two studies thus differ simultaneously in **provenance, instrument, model version, and access path** — none of which is individually identified by comparing their published leaderboards.

3. Methods

3.1 Design: hold queries and answers fixed, vary only the instrument

We reuse Real-POCQi's public artifacts unchanged: the 620 queries, the four systems' verbatim answers, and the blinded human pairwise ratings. On a seed-fixed sample of **150 queries** (`random_state=62`; 600 answers) we administer a **second instrument** — Nature-style absolute 1–4 rubric scoring — to the *same* answers, then derive the *same* OE-vs-rest win-difference metric from both instruments. Because queries, answers, provenance, and model versions are identical across the two instruments, any change in the metric is attributable to the instrument. (For transparency, `random_state=62` was fixed once at the outset and reused for the bootstrap RNG; it coincides with Nature's generation seed but plays no role in our result — the instrument contrast is computed on the same fixed 150-question draw regardless of seed, and the full-bank reproductions in §4.4 match Real-POCQi independently of the subsample. We did not re-draw to obtain a preferred outcome.)

3.2 The rubric instrument (LLM-judge panel)

Each answer is scored 1–4 (1 unacceptable ... 4 excellent) on the five Real-POCQi axes by a blinded panel of four judges — GPT-5.5, Claude Opus 4.8, Grok-4.3, Gemini-3.5-flash — chosen to span vendors and to include contestant families (enabling a self-preference measurement) and one family with no contestant (Grok). We stress that Grok is a *family-neutral* reference, **not a bias-free** one: as an LLM it may still share the panel's general preference for frontier-style prose (structure, breadth, hedging), a house effect that leave-one-judge-out cannot remove (§4.2, Limitation iii). Judges see only the question and one answer, are told nothing about the source system, and return JSON scores only (system prompt in `judge/grade.py`). All judges run at **high reasoning effort**; we **verify** reasoning-token consumption on a real grading item (GPT-5.5 3,071; Grok-4.3 1,880; Gemini-3.5-flash 922; Opus-4.8 443 thinking tokens; `judge/verify_thinking.py`). 2,388 of 2,401 answer-grades completed (0.5% failures — transient timeouts and a few Gemini truncations — balanced across systems).

3.3 Metric and statistics

For each axis we compute the **panel-mean 1–4 score per (question, system)**, then form the OE-vs-rest **win-difference** = 100 × (wins − losses) / comparisons across the three frontier systems per question. This re-expresses the rubric scores through the **same one-vs-rest rule** Real-POCQi reports for human pairwise — a win-ratio-family contrast[22] in the arena-style preference tradition[20] — but the two are **not the identical measurement** (§3.3a), so we treat the win-difference as a common *summary* applied to both instruments, not as an instrument-invariant quantity.

Uncertainty — a crossed question × judge bootstrap. The naïve cluster bootstrap resamples only *questions* and treats the four judges as a fixed population; its CIs answer "how stable is this across questions" but say nothing about "how stable is this across the *choice of judges*" — the dimension a skeptic cares about most here, because three of the four judges are contestant families and GPT-5.5 self-prefers by +0.481 points (§4.2). We therefore make **judges a random factor**: each of 2,000 replicates resamples questions with replacement (cluster on `question_id`) **and** resamples the four judges with replacement, recomputes the panel mean from the resampled judge multiset, and recomputes the win-difference (`judge/bootstrap_panel.py`). Judges are resampled once per replicate and applied consistently to the pairwise and rubric cells (they are the same four judges under both instruments). We report **both** the question-only CI (comparable to prior work) and this wider **crossed** CI. **Primary inference, however, is the aggregation-matched, same-judge format component** (`judge/robust_analysis.py`, §4.5a): because four purposive judges are a *fixed panel* rather than a random sample, we use fixed-judge question-cluster bootstrap CIs as primary and treat the crossed judge-resampling CI as panel-composition *sensitivity*. We also report the format component on the **native 1–4 scale**, under **tie-band** dead-zones (±0.125/0.25/0.5) and panel-median aggregation, with **Holm-adjusted** bootstrap *p*-values and simultaneous **max-|T|** intervals across the five axes, and on **exact common support** (matching every question×opponent×axis for which a human rating exists) so the rater term B−A is not confounded by support (§4.5b–d). Robustness: **leave-one-judge-out** (recompute dropping each judge). Bias: **self-preference** = mean own-family minus mean others, paired within (question, axis). **Inter-judge agreement** = Spearman correlation of per-(question, system) mean scores. Human comparison uses the `qa_text_only` render mode (matching the citation-free OE answers in the dataset).

What the crossed bootstrap changes. Adding judge uncertainty widens every CI, and two previously "significant" per-axis claims do not survive it: the small **source-quality** OE edge (+12.0 pp) moves from a question-only CI that excluded zero [+1.3, +22.0] to a crossed CI that **includes** zero [−8.2, +29.5], and the **clinical-utility** reversal likewise loses significance (crossed CI [−22.4, +8.3]). The **headline accuracy reversal survives**: crossed CI [−37.8, −4.3], still strictly negative. We rely only on claims that survive the crossed CI and explicitly retract the two that do not.

3.3a The win-difference is a construct re-expression, not an identity

A subtle but important point of honesty: cell A's win-difference comes from a **genuine forced choice** — a physician picks A, B, or tie. Cell C's win-difference is **synthesized** by thresholding continuous panel-mean rubric scores (OE panel mean > frontier panel mean ⇒ a "win"). Because the mean of four integer scores is near-continuous, exact ties almost never occur, so essentially every question is forced to a win or a loss on a razor-thin margin — which is precisely why a mean rubric gap of only −0.127 of one point (accuracy) becomes a −29 pp win-difference. The win-difference metric was designed for a paradigm where ties are real (forced choice with a tie option) and we are applying it to one where we have engineered ties away; it is therefore **hypersensitive at the decision boundary**. The honest statement is not "the same metric under two instruments" but "we re-express the rubric means through the same one-vs-rest rule, whose sign is decisive but whose magnitude is boundary-sensitive." We accordingly report both the win-difference *and* the native score gap (§4.1), and lean on the *sign and its crossed-CI significance*, not the raw magnitude.

Multiplicity. We test five axes across three cells; we report per-axis 95% crossed-bootstrap CIs without a formal family-wise correction, so per-axis claims near the null boundary should be read as descriptive. After the crossed bootstrap only two results are individually significant — the **accuracy** reversal (crossed CI [−37.8, −4.3]) and the **instrument-format component** of the decomposition (§4.5, significant on every axis) — and both are far enough from the boundary to survive standard Bonferroni/Holm adjustment across five axes. We flag every borderline axis explicitly rather than over-claiming per-axis significance.

3.4 The pairwise cell (2×2 decomposition)

To separate the instrument from the rater population we administer a **second instrument to the same LLM panel**: blinded forced-choice **pairwise** preference (A/B/tie per axis), OpenEvidence vs each of the three frontier systems, on the same 150-query sample (`judge/pairwise.py`). Slot order is deterministically randomized per (question, opponent, judge) via a hash and de-blinded at scoring, removing position bias. We de-blind each A/B/tie verdict to an OE win/loss/tie and compute the OE-vs-rest win-difference (here a *native* forced choice, not a thresholded score; §3.3a). This yields cell **B = {pairwise, LLM}**, which — with cell A = {pairwise, human} (Real-POCQi) and cell C = {rubric, LLM} (§3.2) — gives three of the four cells of the {pairwise, rubric} × {human, LLM} design and identifies the **rater-modality effect (B−A)** and the **instrument-format effect (C−B)** (§4.5). Judges run at high reasoning effort; all four judges completed 448–450/450 comparisons (1,798/1,800 overall). We bootstrap the decomposition components **jointly** under the same crossed question × judge scheme as §3.3 (resampling questions and judges together across cells A/B/C), so B−A and C−B carry propagated CIs rather than being point-estimate arithmetic. Because cell B already exists in a restricted form inside Real-POCQi itself, we position it as a replication-and-extension, not a wholly new measurement (§4.5a).

4. Results

4.1 The instrument flips the winner (Figure 1, Table 1)

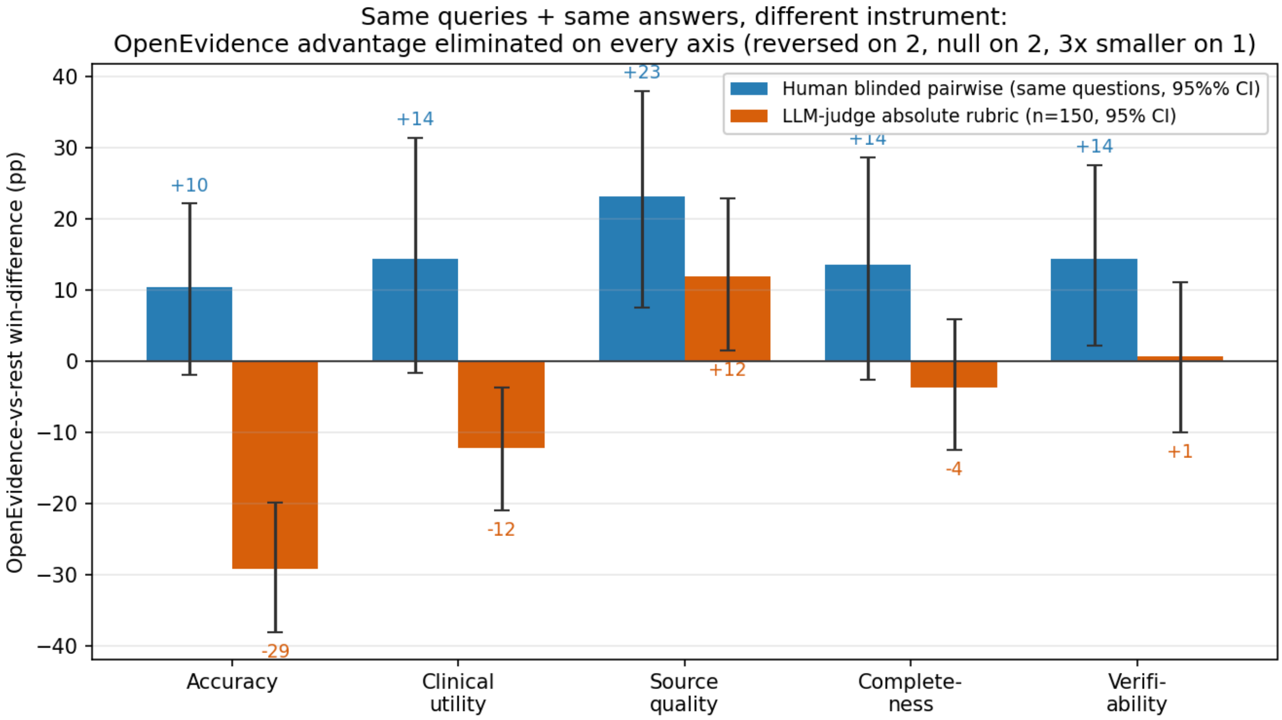

Figure 1. Instrument existence proof. OE-vs-rest win-difference (percentage points) per axis under the human blinded pairwise instrument (same 150 questions) versus the LLM-judge absolute-rubric panel (n = 150). Error bars show the question-only cluster bootstrap; the wider **crossed question × judge** CIs that we treat as primary are given in Table 1. Holding queries and answers fixed and changing only the instrument eliminates OE's advantage on every axis; under the crossed CI the effect is individually significant on accuracy (a sign reversal) and directionally consistent but not per-axis significant on the others.

Table 1. OE-vs-rest win-difference (pp) by axis under each instrument, **both computed on the same 150-query sample**. For the LLM-rubric cell we give both the question-only cluster bootstrap (comparable to prior work) and the wider **crossed question × judge** CI (§3.3), which we treat as primary. The human column is the human pairwise win-difference restricted to the identical questions (86–119 of 150 carry human text-only ratings, by axis); the "full data" column is the win-difference on Real-POCQi's complete text-only bank (reproduced in §4.4), shown for reference because the subsample is underpowered. The native-gap column gives the OE-minus-mean-of-frontier gap on the raw 1–4 rubric (see scale note).

Axis	Human pairwise, same 150 [95% CI]	Human, full data (ref)	LLM rubric, Q-only CI	LLM rubric, crossed q×judge CI	Native gap (pts)	Verdict (crossed CI)
Accuracy	+10.4 [−1.8, +22.2]	+24.4	−29.1 [−37.8, −20.0]	**−29.1 [−37.8, −4.3]**	−0.127	sign flips; **negative, survives judge resampling**
Clinical utility	+14.4 [−1.6, +31.4]	+29.5	−12.2 [−21.1, −3.3]	−12.2 [−22.4, +8.3]	−0.021	sign flips, but **not sig under crossed CI**
Source quality	+23.2 [+7.6, +38.0]	+38.1	+12.0 [+1.3, +22.0]	+12.0 [−8.2, +29.5]	+0.103	attenuated ~3×; **not sig under crossed CI**
Completeness	+13.6 [−2.5, +28.7]	+30.3	−3.6 [−12.7, +5.8]	−3.6 [−14.4, +12.7]	+0.004	collapses to null
Verifiability	+14.4 [+2.2, +27.6]	+25.5	+0.7 [−9.6, +11.1]	+0.7 [−13.6, +13.8]	+0.003	collapses to null (see verifiability caveat, §4.3)

The headline is accuracy. Real-POCQi's central finding is that physicians prefer OE on accuracy (+24.4 pp on the full data; +10.4 pp, CI −1.8 to +22.2, on this 150-query subsample — same sign, but the subsample carries only 86 accuracy ratings and is underpowered). On the **same answers**, the LLM-rubric panel scores OE **−29.1 pp**, and — crucially — this stays significantly negative under the crossed question × judge bootstrap **[−37.8, −4.3]**: the reversal is not an artifact of which judges we happened to pick. So the instrument swap moves the accuracy verdict from OE-favoring (significantly so at full power; positive but CI-crossing in this subsample) to *significantly OE-disfavoring*.

Two per-axis claims from the earlier draft do not survive judge uncertainty, and we retract them. Once judges are treated as a random factor, the **clinical-utility** reversal (crossed CI [−22.4, +8.3]) and the small residual **source-quality** OE edge (crossed CI [−8.2, +29.5]) both cross zero. The honest picture is therefore narrower than "reversed on two, positive on one": under the crossed CI, **accuracy is the one axis with a significant instrument-driven reversal**, and the other four are individually indistinguishable from null. This is a weaker per-axis result but a more defensible one — and the decomposition (§4.5) recovers a stronger, judge-robust pattern at the *component* level. We deliberately avoid framing this as "a significant +24 becomes a significant −29": at the existence-proof scale the human accuracy estimate is not itself significant, so the clean claim is a **sign reversal to a significantly-negative rubric verdict on identical content, robust to judge resampling**, corroborated by the full-data human sign.

Note on scale (native vs win-difference). The win-difference is a sign-of-the-gap statistic: because per-question frontier scores cluster tightly, a mean rubric gap as small as **−0.127 of one point** (accuracy) is enough to flip the majority of head-to-head comparisons and produce a −29 pp win-difference, while a near-zero gap (completeness +0.004, verifiability +0.003) yields a null win-difference. We therefore report both scales: the **native gap** shows the *effect size is modest in absolute terms*, and the **win-difference** shows it is *directionally decisive* under the same one-vs-rest rule Real-POCQi used. Neither instrument's metric should be read as a large clinical-quality gulf; the point is that the two instruments order the systems differently on identical content.

4.2 The reversal is not merely a self-scoring artifact

LLM-as-judge panels are now a standard evaluation tool[23] but are known to favor their own generations,[24] so we measure this directly. Contestant-family judges self-preferred (GPT-5.5 **+0.481**, Opus **+0.121**, Gemini **+0.004** points, own family minus others). Yet leave-one-judge-out shows the **accuracy reversal survives dropping any single judge, including GPT-5.5** (−12.3 pp with GPT-5.5 removed — still negative). GPT-5.5 amplifies the flip roughly fourfold but does not create it. By contrast, the clinical-utility reversal is GPT-

5.5-dependent (→ +2.5 pp without it), consistent with the crossed-CI finding that clinical utility is not judge-robust (§4.1).

Two honest caveats about the panel itself. *(i) The Gemini seat is flash-tier, not Pro.* Our Gemini judge is `gemini-3.5-flash` — a smaller, cheaper, weaker evaluator than the `gemini-3.1-pro` model that is one of the *contestants*, and it is the seat that stalled on quota mid-run (§4.5). Its self-preference of **+0.004** should therefore *not* be read as evidence that the Gemini family is unbiased: a flash-tier judge may simply be a poor discriminator, registering little preference of any kind. Using flash while the contestant is Pro is an apples-to-oranges asymmetry; re-running the Gemini seat at Pro tier and high reasoning is a stated follow-up. *(ii) Grok is a single neutral anchor.* Our cleanest causal handle — that the pro-frontier tilt is instrument-specific because the one family-neutral judge (Grok) still prefers OE under pairwise (§4.5) — rests on **n = 1 non-contestant model**. One neutral judge cannot separate a true instrument effect from Grok's own idiosyncrasies; a second genuinely neutral judge (e.g. DeepSeek, Mistral, Qwen, or Llama) would convert this from a single data point into a pattern, and we flag its absence as a real limitation rather than leaning on Grok as if it were a clean control.

A residual house effect remains, however. Leave-one-judge-out and the self-preference statistic only address *family-specific* bias — a judge favoring its own vendor's answer. They do **not** remove the bias all four LLM judges may *share*: a common preference for the structured, comprehensive, frontier-style prose that general-purpose LLMs produce (and are trained on), against OE's terser clinician-tuned framing. Because even our family-neutral judge (Grok) is itself an LLM, no judge in the panel is a clean control for this shared stylistic prior. The frontier "win" under the rubric is therefore best read as *"frontier answers score higher on an LLM-administered rubric,"* not as a family-free verdict of clinical superiority — a caution we carry into the Discussion and Limitation (iii). The LLM-pairwise cell (§4.5) directly tests this house effect and finds it is **specific to the rubric instrument**: the same judges — including the family-neutral one — prefer OE under pairwise. Human- administered rubric scoring would be the remaining control (pre-registered, §7).

4.3 The instrument is noisy — and one axis is barely measurable in either study

Inter-judge Spearman agreement was modest (0.19–0.47), closely mirroring the low item-level agreement the Nature study reports for its own human raters ($\alpha \approx 0.10$–0.20). Rubric scoring — whether by humans or LLMs — is a higher-variance instrument than forced-choice preference; this is part of *why* it can diverge from pairwise, not a defect unique to our panel.

A specific caution on verifiability. Real-POCQi reports that its own physicians' weighted Cohen's κ was 23–38% on four axes but only **9% on verifiability** — essentially chance agreement. Verifiability is therefore an axis that *neither instrument reliably measures*: when our rubric cell "collapses to null" on verifiability, that is unremarkable, because the human pairwise instrument cannot resolve it either. We accordingly **drop verifiability from any headline framing** and treat its collapse as uninformative rather than as evidence of an instrument effect; the same discount applies, more softly, to source quality. This strengthens the general "both instruments are noisy" point while removing any per-axis weight we might otherwise have placed on verifiability.

4.4 Public-data reproductions (pipeline validation)

Our one-vs-rest win-difference on the human text-only data reproduces Real-POCQi to <1 pp on every axis (e.g., accuracy +24.4 vs 24.7 published). A **proper within-question paired** citation-halo analysis (`judge/robust_supplementary.py`; the earlier `analysis/analyze.py` version compared two independent groups restricted to a common question set, which is not a paired test — reviewer-flagged and corrected) uses the 80 questions rated in **both** render modes and computes the within-question difference in OE preference (with-citations minus text-only): **+0.14 on the −1...+1 preference scale (44 of 76 decisive questions positive)**. So showing citations does raise OE's preference within the *same* question — a **modest but genuine halo**, not the near-zero "selection artifact" the earlier draft reported — which is consistent with citations being one channel of OE's pairwise advantage and motivates the randomized citation arm in §7. Measured question-level SD = 0.598 grounds the power analysis.

4.5 Decomposing the swing: a three-cell path decomposition (Figure 2, Tables 2, 2a)

A natural objection to §4.1 is that swapping {human pairwise} → {LLM rubric} changes **two** things at once — the *rater population* (physicians → LLMs) and the *evaluation format* (forced-choice preference → absolute rubric) — so the reversal could be a human-vs-LLM effect rather than a format effect. The two source studies share exactly this confound. We address it in two steps. First (§4.5a, our **primary** analysis) we isolate the format effect by holding the *rater* fixed: the **same** LLM judge scores the same answers under both formats. Second (§4.5b) we fill three cells of the 2×2 {pairwise, rubric} × {human, LLM} design — A = {pairwise, human} (Real-POCQi), B = {pairwise, LLM} (new; the same four-judge panel, blinded, high reasoning effort, `judge/pairwise.py`), C = {rubric, LLM} (§4.1) — and decompose the human-pairwise→LLM-rubric swing into a **rater component (B−A)** and a **format component (C−B)**. This is a **three-cell *path* decomposition, not a completed factorial**: the fourth cell {human, rubric} is missing, so we do not identify the format effect among humans or the rater×format interaction (§8).

4.5a Primary: the format effect with the rater held fixed (aggregation-matched)

The assumption-light test of "does the format alone move the ranking" holds the rater fixed. For each of the four judges we convert **that judge's own** rubric scores into an OE-vs-frontier win/tie/loss (thresholding its score gap) and pool them exactly as **that same judge's** pairwise votes are pooled, then average the per-judge format change (C_j - B_j) (`judge/robust_analysis.py`). This also removes an aggregation mismatch that inflates the panel-level numbers in §4.5b — there, cell C averages four judges' scores **and then** thresholds, whereas cell B pools individual votes; the averaging-then-thresholding step alone more than doubles the accuracy reversal (panel −29.1 pp vs individual-matched −13.0 pp; §4.5c).

Table 2a. Aggregation-matched, same-judge **format component** = mean(_j)((C_j-B_j)), OE-vs-frontier, on matched (question, opponent) support (n = 391 question×opponent units). Question-cluster bootstrap CIs (fixed-judge primary — four purposive judges are a fixed panel, not a random sample); Holm-adjusted two-sided *p* across the five axes; simultaneous max-|T| CIs. Every axis is sign-consistent across **all four** judges individually.

Axis	Format effect (pp)	95% CI (fixed-judge)	Simultaneous CI	Holm *p*	Sign-consistent (4/4 judges)
Accuracy	−16.9	[−22.3, −11.4]	[−23.9, −9.9]	0.003	✓
Clinical utility	−21.3	[−28.1, −14.3]	[−30.2, −12.4]	0.003	✓
Source quality	−40.6	[−45.8, −34.9]	[−47.6, −33.6]	0.003	✓
Completeness	−37.4	[−43.9, −30.1]	[−46.2, −28.6]	0.003	✓
Verifiability	−45.8	[−51.5, −40.3]	[−53.1, −38.5]	0.003	✓

This is the paper's robust core. With the rater held fixed, switching from pairwise preference to absolute rubric drives the OE-vs-frontier verdict negative on **all five axes**, sign-consistent across **every** judge, surviving simultaneous inference across the five axes, robust to resampling the four judges as a random factor (crossed CIs still exclude zero, e.g. accuracy [−29.7, −4.3]) and to removing OE's citation-bearing answers (citation-free accuracy −17.8 pp; §4.5d). The effect is **largest on the evidence-presentation axes** (source quality −40.6, verifiability −45.8) and **smallest on accuracy** — the reverse of the earlier draft's "marquee accuracy" framing. On the native 1–4 scale the OE–frontier rubric gap is only −0.125 on accuracy (CI [−0.2, −0.1]) and near zero on the content axes; the format nonetheless reverses the ranking because a forced pairwise choice amplifies the small, consistent evidence-presentation advantages (citations, clinician-tuned framing) that a ceiling-bound absolute rubric compresses (mechanism: §6, Supplementary).

Cell B is a replication-and-extension, not a wholly new experiment. Real-POCQi already ran an LLM-judge *pairwise* experiment (their §2.4 / Methods §5.7): they took the same questions, answers, and pairs the physicians saw, blinded and position-randomized them, and had GPT-5.5, Gemini 3.1 Pro, and Claude Opus 4.8 grade them on the same five axes.[17] Our cell B **replicates that design and extends it in three ways** — a fourth, *family-neutral* judge (Grok), **high** reasoning effort (they ran at minimal thinking), and propagated crossed-judge CIs. This matters two ways. First, we credit them: we are not the first to administer an LLM pairwise instrument here. Second, it *strengthens* cell B — because their judges ran at minimal thinking and ours at high, cell B reproducing across both reasoning regimes is a genuine robustness point, and their released ratings could in principle be used to triangulate cell B directly. Their headline is concordant with ours: LLM judges **agreed with human experts on which system was best** (i.e. OE at the one-vs-rest winner level) while disagreeing on the lower ranks — exactly the pattern our OvR win-difference is built to surface (see the rank-resolution caveat below).

One component of B−A is a genuine rater confound, not pure "modality," and it lands on accuracy. Real-POCQi **specialty-matched** each physician grader to the question topic and argues this matching optimizes evaluation on the *accuracy* axis specifically.[17] Our LLM judges are generalists with no such matching. So B−A on accuracy conflates the pairwise→pairwise rater-*modality* change with a loss of **specialty expertise** — a confound that would make the human cell A *more* accurate-sensitive. This is a caveat on the matched accuracy rater term (−7.9 pp, non-significant; §4.5b): if specialty-matching inflates cell A's accuracy edge, the true human→LLM accuracy change could be somewhat more negative than measured — but as estimated it is not distinguishable from zero, and we make no claim that it is large. We name specialty-matching explicitly as part of B−A rather than burying it in "protocol differences."

Rank-resolution caveat: cell B's agreement lives in the winner-friendly OvR regime. The concordance we credit — LLM judges reproducing the human pairwise winner — must be read against what Real-POCQi actually found about LLM-judge *ranking* fidelity: their LLM judges agreed with physicians on **which single system was best** but their rank correlations across the *full* ordering were low-to-negative (e.g. Kendall $\tau \approx -0.200$ for GPT-5.5 and ≈ -0.067 for the jury against the human ranking).[17] The one-vs-rest win-difference is by construction a **winner-resolving, not a rank-resolving** contrast: it asks only "is OE preferred to the frontier field," which is exactly the question on which LLM and human judges *do* concur, and it is deliberately blind to the lower-rank disagreements where they diverge. So cell B's reproduction of cell A is genuine but should be understood as agreement *at the resolution our metric operates on* — the top of the order — not as evidence that LLM judges recover the physicians' complete system ranking. This cuts both ways for us: it is why cell B is a fair test of the "do LLMs disagree with physicians about OE" question (they do not, at winner level), and why we do not over-read cell B as a general validation of LLM-judge pairwise scoring.

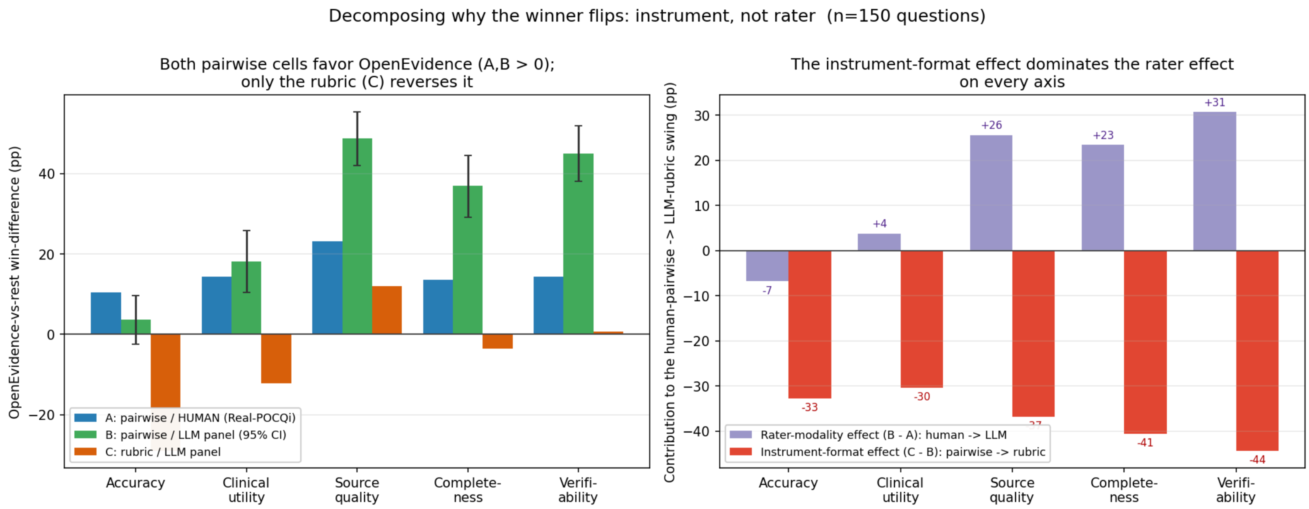

Figure 2. The three-cell path decomposition (n = 150), panel-level. *Left:* three cells of the {pairwise, rubric} × {human, LLM} design per axis — both pairwise cells (A: human; B: LLM) favor OpenEvidence; only the rubric cell (C) reverses it. *Right:* the human-pairwise→LLM-rubric swing split into a rater component (B−A) and a format component (C−B); the format component is negative with a crossed CI excluding zero on **every** axis. Note: this panel-level view compares cells on non-identical supports; the **exact common-support** decomposition (identical keys) is Table 2b, and the robust, rater-fixed format estimate is Table 2a. On matched support the accuracy rater term is small and non-significant (−7.9 pp, p=0.30); the rater term is significantly *positive* on the evidence axes.

Table 2. Panel-level OE-vs-rest win-difference (pp) across three cells of the 2×2, and the decomposition of the human-pairwise→LLM-rubric swing **with propagated crossed question × judge 95% CIs** on the two components (§3.3; `judge/bootstrap_panel.py`). **Cell A is computed on the same 150-query sample as B and C** (human text-only ratings restricted to those questions), *not* on the full text-only bank — otherwise B−A would absorb a sample-composition difference into the "rater" term. Cell B: n=150; 8,990 axis-verdicts (1,798/1,800 comparisons, all four judges 448–450/450).

Axis	A: pw/human	B: pw/LLM	C: rubric/LLM	Rater (B−A) [crossed CI]	Instrument (C−B) [crossed CI]
Accuracy	+10.4	+3.6	−29.1	−6.8 [−27.5, +14.3]	**−32.7 [−42.8, −8.8]**
Clinical utility	+14.4	+18.2	−12.2	+3.8 [−22.3, +31.9]	**−30.5 [−42.6, −9.0]**
Source quality	+23.2	+48.8	+12.0	+25.6 [−3.4, +54.8]	**−36.8 [−56.9, −22.4]**
Completeness	+13.6	+37.0	−3.6	+23.4 [+2.2, +45.4]	**−40.5 [−53.6, −22.0]**
Verifiability	+14.4	+45.1	+0.7	+30.7 [+7.9, +53.6]	**−44.4 [−62.0, −29.1]**

4.5b Exact common-support decomposition, and what the rater actually contributes

The panel-level Table 2 compares A, B, and C on **overlapping but non-identical** supports: human text-only ratings are sparse (86–119 of 150 questions per axis, sparser per question×opponent), so B−A can absorb a support difference. We therefore recompute the decomposition on the **exact common support** — every (question, opponent) carrying a human vote **and** complete four-judge pairwise **and** complete four-judge OE- and opponent-rubric scores (108 question×opponent units/axis) — so A, B, C use **identical keys**, with a question-cluster bootstrap over the matched clusters only (`judge/robust_analysis.py`).

Table 2b. Exact common-support decomposition (108 question×opponent units/axis; A/B/C on identical keys). B−A is a **protocol contrast** (rater population *and* text-only→Markdown rendering), not a pure rater effect; p is a two-sided bootstrap **tail probability** (inference rests on the CIs). Equal-key–weighted values (each key once) are in the text and move all estimates <3 pp.

Axis	**A: human [95% CI]**	**B: LLM pw**	**C: LLM rubric (indiv)**	**Protocol B−A [CI] (tail p)**	**Format (C−B) [CI]**
Accuracy	+9.0 [−3.3, +21.1]	+1.2	−16.7	−7.9 [−23.5, +7.1] (0.30)	**−17.8 [−27.0, −8.4]**
Clinical utility	+13.1 [−2.6, +30.6]	+15.3	−5.1	+2.2 [−17.5, +21.4] (0.82)	**−20.4 [−31.6, −8.8]**
Source quality	+22.1 [+6.7, +37.7]	+47.2	+7.2	+25.1 [+6.9, +43.0] (0.006)	**−40.0 [−48.8, −31.3]**
Completeness	+12.3 [−3.9, +28.6]	+29.9	−2.8	+17.6 [+1.8, +33.6] (0.034)	**−32.6 [−44.6, −20.9]**
Verifiability	+13.9 [+0.0, +26.7]	+44.0	+0.9	+30.0 [+12.9, +46.2] (0.003)	**−43.1 [−52.2, −33.1]**

Two things follow — and they **correct an error in the previous revision**.

The format component is the robust negative driver, on all five axes. On identical keys, C−B is negative with a CI excluding zero everywhere (−17.8 to −43.1 pp), concordant with the rater-fixed primary (§4.5a). This is the claim we stand behind.

B−A is a protocol contrast (rater population *and* rendering), null on accuracy and *positive* on the evidence axes — not "large and negative." We stress that B−A is not a clean rater-population effect: cell A (human) used the `qa_text_only` render while cells B/C (LLM) scored `answer_markdown`, which carries citations for ~29% of OE answers, so B−A confounds the human→LLM rater change with a text-only→Markdown rendering change (and with Real-POCQi's specialty-matching). A prior revision reported an accuracy B−A of **−24.0 pp** and used it to "retract 'instrument, not the rater'." **That −24.0 was an analysis error** — a support filter was inadvertently vacuous, so cell A was the *full-bank* human edge (+24.4) compared against the *matched* LLM value. On the true common support the accuracy contrast is **−7.9 pp (CI −23.5 to +7.1, bootstrap tail p=0.30): not significant**, essentially the original panel-level −6.8, so it does **not** explain the accuracy reversal. Where B−A *is* significant it points the other way: on source quality, completeness and verifiability it is **positive** (+25.1, +17.6, +30.0) — LLM judges scoring citation-bearing Markdown are *more* OE-favorable than physicians scoring text-only. Tellingly, these are exactly the axes most sensitive to citation presence, which is why we read B−A as a rater+rendering protocol contrast rather than evidence about rater population alone.

Equal-key sensitivity. Because cell A pools human votes (122 across the 108 keys) while B/C weight each key uniformly through their four-judge panels, we re-ran the decomposition giving **every question×opponent key equal weight** (votes averaged within key, bootstrap by question). The numbers barely move: accuracy B−A −5.8 (still NS), source quality +28.7, completeness +18.8, verifiability +29.5 (all still significant), and the format component C−B is **unchanged** (it does not involve cell A). The conclusions are not an artifact of unequal human rating density.

Net position. We do **not** claim "the instrument, not the rater" (B−A is a real, significant contributor on three axes), nor that it is large and negative on accuracy (it is null). Both the format change and the human→LLM protocol contrast matter, but the **format** component — negative on all five axes and established with the rater and rendering held fixed (§4.5a) — is what carries the reversal. The earlier "~83% instrument share" is withdrawn as ill-posed.

4.5c Aggregation and tie-margin sensitivity

The panel-mean-then-threshold rule used for cell C is hypersensitive at the decision boundary. We report each dead-zone at **both** levels — the cell-C win-difference and the format effect C−B — so the two are not confused. On accuracy, a symmetric tie-band of ±0.25 on the panel-mean gap cuts the cell-C win-difference from −29.1 to −13.8 pp (C−B from −30.4 to −15.1; ±0.5 → C −6.7 / C−B −8.0; panel-**median** → C −14.7 / C−B −16.0; individual-then-aggregate format effect −16.9) — all converging on the aggregation-matched −16.9 pp of §4.5a and confirming the −29.1 figure was an aggregation artifact. The evidence-presentation axes are insensitive to the same perturbation at the cell-C level (verifiability stays ≈ 0), because their *format* effect lives in C−B, not in C alone; so the demotion of accuracy is specific to that axis, not a general softening (`judge/robust_analysis.py`, `tie_margin_sensitivity`).

4.5d Rendering, opponent-specificity, and multiplicity

The format effect of §4.5a is **not** an artifact of OE's citations: OE's `answer_markdown` carries citations in 29% of answers (frontier models 0.2%), but **restricting the same-judge analysis to the question set whose OE answer contains no detected citation markers** (a subset restriction — we do not strip citations, and this does not equate the human `qa_text_only` and LLM `answer_markdown` renderings) barely moves the format component (accuracy −17.8 vs −16.9; all axes within 5 pp). It is also **not driven by a single opponent**: the accuracy format effect is −7.9 (vs GPT), −21.5 (vs Claude), −21.3 (vs Gemini) — in fact *smallest* against GPT, because GPT's answers are down-rated by LLM pairwise and rubric alike. On **multiplicity**, we report Holm-adjusted **bootstrap tail probabilities** (two-sided, resampling-based, not a formal null-randomization test) and simultaneous max-|T| intervals for the primary format component (Table 2a); primary inference rests on the intervals, with the tail probabilities as a secondary summary. Every axis survives both, so the "negative on all five axes" claim is robust to family-wise correction rather than merely asserted to be.

This also sharpens the house-effect discussion (§4.2). Under the pairwise instrument, the **family-neutral judge (Grok) prefers OE on all five axes** (+11 to +65 win-diff) and the contestant **GPT-5.5 is the *least* OE-favorable** judge — the *opposite* of what a shared pro-frontier stylistic bias would predict. The pro-frontier tilt is therefore specific to the **rubric** instrument, not a blanket LLM prejudice against OE. (One robustness note: Gemini-3.5-flash initially stalled at 297/450 pairwise verdicts when the Google API account exhausted its credit quota mid-run; after credits were restored we completed the cell to 448/450, and the point estimates shifted <3 pp from the partial-coverage run — i.e., the conclusion never depended on the missing data.)

4.6 The rubric reversal is not a length artifact (Table 3)

Answer length is the most-cited confound in both source studies. A precise word on how it is handled: Real-POCQi does not leave length wholly uncontrolled — it **reports and stratifies** by length (their Table D3 / Fig D6, OE ≈ 3,584 vs Gemini ≈ 3,516 characters) but does not **normalize** generation to equal length; so the accurate statement is "length is stratified and reported, not equalized," not "uncontrolled." The natural worry is that the rubric rewards longer, more comprehensive answers. We test this on the rubric cell without regenerating any answers (`judge/length_analysis.py`). The premise fails at the first step: **OpenEvidence is among the *longest* providers** — median 3,600 chars, essentially **tied with Gemini (3,586)** and well above Opus (3,294) and GPT-5.5 (2,232) — so OE is decisively longer only than GPT, but it is never the *short* answer, and it still *loses* the accuracy rubric. A "longer answers win" mechanism therefore runs *backwards* against the finding: the co-longest provider loses. We confirm this three ways (Table 3): (i) the per-axis length **slope** is tiny (−0.015 to +0.042 rubric points per 1,000 characters); (ii) the **length-adjusted intercept** — the expected OE-minus-frontier score gap at *equal length* — is statistically indistinguishable from the raw gap on every axis (accuracy −0.12 [−0.17, −0.08] adjusted vs −0.127 raw); and (iii) the accuracy win-difference stays negative **whether OE's answer is longer (−34.3) or shorter (−19.9) than its opponent's** — OE loses on accuracy even when it is the longer answer.

Table 3. Length sensitivity of the rubric cell (n=150; 2,000-replicate cluster bootstrap). Native-scale OE-minus-frontier score gap: raw vs length-adjusted (intercept at equal length); and the accuracy win-difference split by whether OE is the longer or shorter answer.

Axis	Raw gap (pts)	Length-adjusted gap [95% CI]	Slope (pts / +1k chars)	Win-diff: OE longer	Win-diff: OE shorter
Accuracy	−0.127	−0.12 [−0.17, −0.08]	−0.015	−34.3	−19.9
Clinical utility	−0.021	−0.03 [−0.06, +0.01]	+0.016	−14.1	−8.4
Source quality	+0.103	+0.09 [+0.03, +0.15]	+0.042	+15.9	+6.0
Completeness	+0.004	−0.01 [−0.05, +0.03]	+0.039	+0.7	−10.8
Verifiability	+0.003	−0.01 [−0.06, +0.05]	+0.025	+1.4	+0.0

Length has a small positive association with source-quality/completeness/verifiability scores (slopes +0.03 to +0.04 per 1,000 chars) but essentially none with accuracy, and adjusting for it leaves every axis's OE-vs-frontier gap unchanged within CI. The accuracy reversal is therefore **not** explained by length. (This does not make length irrelevant in general — a fully length-matched *generation* study is still pre-registered, §7 — but it removes length as an alternative explanation for the observed reversal.)

5. Critiques of both source studies

The critiques below are **independent of the instrument existence proof** (§4): they hold whether or not the instrument drives the disagreement, and they apply symmetrically to both source studies. We include them because they bound how far *either* published leaderboard can be trusted for procurement, but a reader interested only in the instrument result can treat this section as standalone context. Full detail in `CRITIQUES.md`. Points (a)–(b) concern the systems and settings compared; points (c)–(e) concern the instruments themselves and bear directly on interpretation:

(a) The comparator set omits the AI clinicians actually use. Both studies benchmark **bare frontier API models**; neither includes the **ChatGPT product** (consumer or clinician deployment), whose system prompt, retrieval/browsing, memory, and safety guardrails move the very axes under test. This is especially asymmetric because OE is itself a curated clinical *product*; Nature further queries clinical tools via browser but frontier models via API. A fair benchmark should include a **product arm** distinct from the raw model endpoint.

(b) Reasoning effort is unreported or uncontrolled. Nature reports temperature 0.0 and a seed but **never states reasoning effort** (its cost table confirms reasoning tokens were spent, at an unspecified level); Real-POCQi reports only that "thinking was automatically determined by the LLM." These are two *distinct* undisclosed knobs — whether extended thinking is **on at all**, and if so at what **level (high/medium/low or an explicit token budget)** — and reasoning effort is the largest controllable performance lever for these models, able to swing double digits on exactly the benchmarks in play. Two consequences follow. *Across studies*, neither head-to-head is reproducible on this axis, and their absolute numbers are not comparable. *Within* a study, "automatic" or unstated effort is worse than merely non-reproducible: if a study's thinking setting is auto-determined per query or per model, it may **not be held constant across the very systems it is comparing** — so even that study's internal ranking may reflect an uncontrolled effort difference rather than a capability difference. (Note, too, that temperature 0.0 is largely inert for reasoning models, which sample the reasoning trace regardless; reporting it can create a false impression of determinism.) Our study instead pins effort = high for all four judges and **verifies** token consumption on the real task (§3.2), so the instrument comparison here is not vulnerable to this confound.

(c) The winning margin sits inside the rubric's own noise floor, and the 1–4 scale compresses it. Two linked problems undercut how decisively Nature's rubric separates the systems. *First, a noise-floor* *problem.* On the real-clinical-question benchmark the frontier systems finish at Gemini 3.62 / GPT 3.54 / Claude 3.52 — a spread of **0.10 of one point across the top three** — with OE at 3.24, i.e. roughly 0.3 point below an essentially indistinguishable frontier cluster, on n = 100 questions graded at Krippendorff α ≈ 0.10–0.20. At that agreement level and sample size the between-frontier ordering is within measurement noise (the "winner" among Gemini/GPT/Claude is not robustly identified), and even the OE gap is a fraction of a rubric point; the study's own low α makes the fine-grained leaderboard order fragile. *Second, a ceiling/compression problem that is also a mechanism for our reversal.* A 1–4 integer rubric has almost no headroom for strong answers — competent clinical responses pile up at 3–4, so the instrument cannot express *how much* better one good answer is than another, and small stylistic preferences (structure, breadth, hedging) get magnified into rank order because the usable dynamic range is one or two points. This compression is exactly why a mean gap of −0.127 point becomes a −29 pp win-difference in our cell C (§3.3a): the rubric floor/ceiling turns near-ties into decisive-looking orderings. A coarse absolute rubric is therefore a **low-resolution instrument for near-parity systems**, and both the Nature leaderboard and our rubric re-expression inherit that low resolution. This is not a reason to prefer pairwise uncritically — §5(d) levels a symmetric charge at the preference instrument — but it bounds how much weight the Nature rubric ordering can carry.

(d) The pairwise instrument has symmetric, opposite biases — it is not the "true" instrument either. Nothing here should be read as vindicating forced-choice preference as ground truth. Pairwise preference is known to reward surface features — answer length, formatting, citation presence, confident tone — independently of correctness, and Real-POCQi's own data show the mechanism: on the 80 questions rated in **both** render modes, a **proper within-question paired** test finds that showing citations raises OE's preference by **+0.14 on the −1…+1 scale** (44 of 76 decisive questions positive; §4.4) — a modest but genuine citation halo *within the same question*, i.e. part of the measured preference edge tracks the *presence of citations* rather than answer accuracy. (This corrects the earlier draft, which used an independent-groups test on a common question set and wrongly reported the halo as vanishing.) A preference instrument that can be moved by attaching references is construct-contaminated, in the opposite direction, as a compressed rubric. The honest position is that **neither instrument is a clean measure of clinical quality**: the rubric over-rewards frontier-style breadth and compresses near-ties; pairwise over-rewards citations and verbosity. Our claim is the *relative* one — that switching between these two biased instruments is sufficient to flip the winner — not that either endpoint is correct.

(e) The five rubric axes are not independent, so "reversed on k axes" overstates the evidence. Source quality and verifiability both essentially ask *"are there citations, and are they real,"* and accuracy, completeness, and clinical utility are strongly co-scored; the axes are near-collinear rather than five independent measurements. Two consequences: per-axis "significance" counts should not be read as five independent confirmations (a single latent "cited, thorough, frontier-style" factor drives much of the axis structure in both instruments), and this non-independence is *why* we lean on the accuracy axis and the decomposition rather than tallying axes. It also tempers Real-POCQi's own multi-axis sweep: OE winning "on all five axes" is closer to winning on **one or two underlying constructs** measured five ways. We treat the axis set as a small number of correlated constructs throughout, not as five degrees of freedom.

(f) Both studies' data pipelines are run by an interested party, with symmetric — not one-sided — confounds. We have stressed the provenance symmetry (each sourced queries from the platform that won); three further data-pipeline facts deserve naming because they cut in *different* directions and a fair reconciliation must not weaponize only the ones that suit its thesis. *(i) End-to-end control by the winner.* Real-POCQi's pipeline was run by OpenEvidence: OE selected the query pool (≈3,600 → 620 after filtering) from its **own** platform traffic, paraphrased queries via an LLM (Opus 4.6) for de- identification, and paid the physician raters — a chain in which the party under study controls sampling, wording, and rater incentives end to end. Nature's RCQ pipeline is analogously insider-run (NYU Langone's own GPT deployment). Neither is neutral. *(ii) A rater-familiarity confound that actually supports the instrument thesis.* Roughly **half of Real-POCQi's raters were themselves OpenEvidence users**, and the reported OE accuracy edge is

larger among OE-users (≈+33.8 pp) than non-users (≈+22.6 pp) — a habituation/familiarity gradient that inflates the human-pairwise cell (cell A) for reasons unrelated to answer correctness. This *strengthens* our reading: part of cell A's OE edge is rater familiarity with OE's house format, exactly the kind of preference-instrument artifact that an absolute rubric does not reward. *(iii) Very low response and completion rates.* Real-POCQi's physician recruitment yielded low single-digit response and completion fractions (on the order of ~1–2%), so the rating panel is a self-selected slice of clinicians; combined with (ii), cell A's preference signal carries a selection/familiarity component we cannot remove from public data. None of this shows OE's answers are worse — it shows the *human-pairwise* cell is itself confounded, which is why we do not treat cell A as ground truth and lean instead on the *within-panel* instrument decomposition (§4.5).

(g) On the fine-grained order, both leaderboards agree more than they disagree. Stripped to what is *robust*, both studies essentially resolve only the extremes: OE is top under pairwise and GPT-class models are strong under rubric, but the middle of each ranking is within noise (§5c). Real-POCQi's frontier trio is near-parity behind OE; Nature's frontier trio (Gemini 3.62 / GPT 3.54 / Claude 3.52) is near-parity ahead of OE. The genuine, reproducible cross-study disagreement is therefore narrow — **where OE sits relative to a tightly-bunched frontier cluster** — and it is precisely that one placement that the instrument flips. This bounds the stakes: we are not reconciling two wildly different capability orderings, but one contested boundary between OE and a frontier pack that both studies otherwise order similarly.

A corroborating confound: Real-POCQi tested **newer** frontier weights (GPT-5.5, Opus 4.8) than Nature (GPT-5.2, Opus 4.6) and still found them losing on human pairwise — a pattern more consistent with an instrument effect than a model-version effect.

6. Discussion

Holding queries, answers, **and the rater** fixed, the evaluation format alone is sufficient to reverse the central claim of a published clinical-AI benchmark. The rater-fixed analysis (§4.5a) makes this attribution clean: the *same* LLM judge, scoring the *same* answers, moves the OE-vs-frontier verdict negative on all five axes when switched from pairwise preference to absolute rubric (−16.9 to −45.8 pp, sign-consistent across all four judges, robust to simultaneous inference, judge-resampling, and restriction to citation-free OE answers). This is the claim we defend.

What we do and do not claim. We do **not** claim "the instrument, not the rater." On **exact common support** (identical keys for A, B, C) the human→LLM **protocol contrast (B−A)** — which changes rater population *and* answer rendering (text-only vs citation-bearing Markdown) — is **null on accuracy (−7.9 pp, CI −23.5 to +7.1, bootstrap tail p=0.30)** and clinical utility but **significantly positive** on source quality (+25.1), completeness (+17.6) and verifiability (+30.0). Because those are the axes most sensitive to citation presence, we read this as a rater+rendering contrast, not evidence about rater population alone. So B−A is a real contributor, but it does **not** explain the rubric reversal (and on accuracy it is indistinguishable from zero — a prior revision's "−24.0 pp, significant" was an analysis error from an unmatched cell A; §4.5b). The reversal is carried by the **format** component, which is negative on all five axes with the rater *and* rendering held fixed. The decomposition is a **three-cell path decomposition, not a factorial**: with cells A = {pairwise, human}, B = {pairwise, LLM}, C = {rubric, LLM} but not D = {rubric, human}, we identify the human→LLM change under pairwise and the pairwise→rubric change within LLMs, but neither the format effect among humans nor the rater×format interaction. Bridging our within-LLM format effect to the human-rated Nature rubric therefore assumes that interaction is small — plausible but untested, and exactly where interaction is most likely (humans may not share whatever prior an LLM-scored rubric rewards). Filling cell D, even with a small human-rubric sample, is the single highest-value follow-up.

A mechanism is visible in the data. The absolute rubric is **ceiling-bound on the content axes** — 72% of accuracy grades, 82% of clinical-utility grades, and 77% of completeness grades are the top score of 4 (entropy ≈ 0.8–1.0 bits of a possible 2) — so it can barely separate systems on content, and the native OE–frontier accuracy gap is a mere −0.125 on the 1–4 scale. On the **evidence-presentation axes** (source quality, verifiability) the rubric is far more dispersed (only ~21% at ceiling, entropy ≈ 1.5) yet still scores the field similarly, whereas a forced pairwise choice sharply rewards OE's citations and clinician-tuned framing. That is why the format effect is largest and most robust exactly on the evidence axes and smallest on accuracy: the two formats encode different value functions, and the rubric's compression is what a pairwise contest undoes. Accuracy, the earlier draft's marquee axis, is in fact the **weakest** and most aggregation-sensitive instance of the effect (§4.5c).

We are not the first to suspect the instrument — Real-POCQi says so itself. Real-POCQi explicitly hypothesizes that its divergence from rubric-style benchmarks is *instrument-driven*, and cites JudgmentBench[25] — which reports that in a high-expertise professional (legal) domain **head-to-head preference recovers expert judgment more faithfully than absolute rubric scoring** — to argue that its own pairwise design is the more valid one. We credit this: our contribution is not the *idea* that the instrument matters but an **executed, controlled demonstration** of it (queries and answers held fixed) plus a decomposition separating instrument from rater. But we also engage the directional claim honestly, because it cuts against our symmetric "neither instrument is truth" framing (§5d): if JudgmentBench is right that head-to-head is the higher-fidelity instrument in expert domains, then the pairwise (OE-favoring) result deserves *more* evidential weight than the rubric (frontier-favoring) result, and our finding would be not merely "the instruments disagree" but "the more valid instrument favors OE." We do **not** assert this — JudgmentBench's generalization to point-of-care clinical answers is untested, our rubric is LLM- rather than human-administered, and a compressed 1–4 scale is a weak version of rubric scoring — but we flag it as the key open question the missing human-rubric cell (D) would help settle: if physicians administering a rubric still down-rank OE, the disagreement is instrument-intrinsic; if they do not, it localizes to LLM-administered rubrics specifically. Either way, adjudicating *which* instrument is closer to clinical truth (not merely that they differ) is the natural next study, and JudgmentBench gives a prior that pairwise may win that contest.

This does not show OE is worse (or better) than frontier models in the clinic — we have **no adjudicated ground truth** — but it shows that "which system is best" is a function of the measuring instrument, and that pairwise-preference and absolute-rubric instruments encode materially different value functions (preference rewards OE's clinician-tuned framing and citations; rubric scoring rewards frontier models' breadth and structure). It also reframes the "LLM-judge self-preference" worry: the pro-frontier tilt is a property of the **rubric instrument**, not of LLM judges per se, since those same judges favor OE under pairwise — including the family-neutral judge. Benchmarks that do not report the instrument as an experimental factor — and that do not control reasoning effort, comparator product-vs-endpoint status, and length — are under-specified for procurement decisions.

7. Pre-registered extension (proposed design, not an executed result)

The executed existence proof identifies the instrument; a full **2×2×2 provenance × instrument × citations** factorial (protocol in `reconciliation_protocol.md`) estimates each factor's causal share, including a randomized citation-halo arm (motivated by §4.4), a length-matched *generation* sub-study (to confirm causally what §4.6 shows observationally — that length does not drive the reversal), a product-vs-endpoint arm (§5a), and reasoning-effort sweeps (§5b). Power is grounded in the measured question-level SD (0.598): ≈250 questions/cell for a 15 pp interaction, ≈390/cell for 12 pp. Because the Nature RCQ corpus is not public, the provenance arm uses a constructed surrogate LLM-platform query corpus; Real-POCQi is directly reusable. **This surrogate is itself a threat to validity:** a corpus we assemble to *stand in* for NYU's HIPAA-restricted RCQ distribution may not reproduce its topic mix, difficulty, or phrasing, so the provenance-arm estimate is only as trustworthy as the surrogate's fidelity to the true LLM-platform query stream. We therefore treat the provenance factor as the **weakest-identified** arm of the proposed factorial and would pre-register the surrogate-construction procedure and a sensitivity analysis over plausible corpus definitions rather than reporting a single provenance estimate as if the corpus were the real thing.

8. Limitations

(i) **No ground truth** — we show formats disagree, not which is correct. (ii) **Aggregation sensitivity** — the panel-mean-then-threshold win-difference amplifies small native gaps at the decision boundary, most acutely on accuracy (panel −29.1 pp vs aggregation-matched −16.9 pp vs a ±0.25 tie-band −13.8 pp; native OE–frontier gap only −0.125 on the 1–4 scale). We therefore make the **aggregation-matched, same-judge** component (§4.5a) primary and treat accuracy as the weakest, most aggregation-sensitive axis; the evidence-presentation axes are robust to the same perturbation. (iii) **Three-cell path decomposition, not a factorial** — we have cells A = {pairwise, human}, B = {pairwise, LLM}, C = {rubric, LLM} but **not** D = {rubric, human}, so we do not identify the format effect among humans or the rater×format interaction. Bridging our within-LLM format effect to the human-rated Nature rubric assumes that interaction is small — untested, and the likely direction of interaction. On **exact common support** the human→LLM **protocol contrast** B–A (rater population *and* text-only→Markdown rendering) is null on accuracy (−7.9 pp, tail p=0.30) and significantly *positive* on the evidence axes (+18 to +30; robust to equal-key weighting), so we do **not** claim "instrument, not the rater" (the rater is a real contributor), nor that the rater is large and negative on accuracy (a prior revision's −24.0 pp was an unmatched-cell-A error). Filling cell D, even at small n, is the top follow-up. B–A also absorbs Real-POCQi's specialty-matching and minor protocol differences, so it is a rater *change*, not a clean modality effect. (iv) **Rendering not matched across the rater contrast** — human cell A used the `qa_text_only` render, but LLM cells B/C scored `answer_markdown`, which carries citations for **29%** of OE answers (frontier 0.2%); so A vs B/C is not presentation-matched and B–A partly reflects rendering. The **rater-fixed format effect (§4.5a) is unaffected** — B and C saw identical text — and is robust to restricting to citation-free OE answers (§4.5d; a subset restriction, not citation removal); a canonical text-only re-render of B/C is pre-registered. (v) **Four purposive judges** — three from contestant families — support a *fixed-panel* claim, not broad random-judge generalization. We report fixed-judge inference as primary, with per-judge sign-consistency, leave-one-judge-out, and self-preference difference-in-differences; the crossed judge-resampling bootstrap is reported only as **panel-composition sensitivity**. Adding genuinely neutral judges is future work. (vi) **Not Nature's exact rubric** — we score the five Real-POCQi axes with an LLM panel on a 1–4 scale; this is a general absolute-vs-comparative contrast, **not** a transport of Nature's RCQ instrument (clinical correctness/completeness/safety/clarity plus harm and hallucination flags, human-administered with ordinal mixed models). (vii) **Length not normalized** (OE 3.8k vs GPT 2.6k chars) — observationally (§4.6, Supplementary) OE is the longest provider yet loses accuracy and the within-opponent length↔score-gap correlation is ≈ 0, but a length-matched *generation* study is pre-registered to settle it causally. (viii) **House effects** — contestant-family judges self-prefer, but this survives as a difference-in- differences (GPT +0.42 beyond panel consensus) and the pairwise cell shows the pro-frontier tilt is **format-specific** (the same judges favor OE under pairwise, including the family-neutral judge). (ix) **Noisy instrument** — modest rubric inter-judge agreement (Spearman ≈ 0.23–0.39). (x) **No rater-level modeling** — the public ratings table has no rater identifier, so human-side rater random effects cannot be fit. (xi) **LLM judges are not bit-deterministic**; we quantify rather than assume stability.

9. Ethics, conflicts, funding

No human subjects or PHI were used; all human ratings are de-identified public data (CC BY 4.0). No Nature-copyrighted data were redistributed (CC BY-NC-ND). **Conflicts of interest:** The author is the founder of Kinvectum AB, an independent medical-AI research venture. The author declares no financial interest in, and no commercial or consulting relationship with, any of the evaluated systems or their providers (OpenEvidence, OpenAI, Anthropic, xAI, Google, or Wolters Kluwer/UpToDate). For transparency we note the source studies' own disclosed conflicts, symmetrically: Real-POCQi's data collection was run by OpenEvidence (the platform under study, which also won it), and the Nature Medicine senior author reports disclosed industry equity/consulting including **Google** — whose Gemini model is the top scorer on that study's headline RCQ leaderboard. Both disclosures are proper and neither implies misconduct; we flag them only because in each study the disclosed interest overlaps the declared winner, which is exactly the kind of provenance/COI symmetry this reconciliation is built to treat even-handedly. **Funding:** This work was funded by Kinvectum AB, which covered the LLM-judge API costs; the funder had no role in study design, analysis, or the decision to publish. **Author contributions:** K.A. designed the study, wrote the analysis and judging code, performed all analyses, and wrote the manuscript (sole author).

10. Data and code availability

All code and outputs are in this repository; `run_all.sh` reproduces every result end to end (`SKIP_JUDGES=1` for the no-API-key public-data subset). Real-POCQi data: Hugging Face `jjfenglab/Real-POCQi` (CC BY 4.0), fetched by `fetch_data.py` and vendored in `data/`. Judge configuration, reasoning-token verification, grades, bootstrap CIs, and figures are regenerable as documented in `README.md`. A machine-readable **per-item instrument-disagreement export** (`judge/export_disagreement.py` → `judge/out/instrument_disagreement.csv`) gives, for every (question, axis, comparator) triple, the rubric and pairwise winners/margins, the number of contributing judges, judge dispersion, and an `instrument_flip` flag, so readers can inspect *where* reversals concentrate; re-aggregating it reproduces the sign of every reported win-difference (a deterministic unit test pins the flip definition). This is a descriptive audit layer, not adjudicated clinical truth. The **primary robust analyses** (aggregation-matched same-judge format component, native-scale gaps, exact common-support A–B–C, tie-margin and multiplicity, opponent- specific effects, rendering verification) are in `judge/robust_analysis.py` → `judge/out/robust_analysis.json`, with `judge/test_robust_analysis.py` pinning the estimand definitions; **exploratory supplementary** analyses (rubric reliability/resolution, self-preference difference-in-differences, axis factor structure, length with opponent fixed effects, paired citation-halo) are in `judge/robust_supplementary.py` → `judge/out/robust_supplementary.json`. Nature Medicine numbers are cited from the published article (s41591-026-04431-5); its RCQ corpus is not public (IRB i23-00510).

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Citations appear as numbered footnotes at the point of use and are collected in full below. Numbering follows order of first appearance (Vancouver style); author lists of six or more are abbreviated with *et al.* per Vancouver convention.

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